

Q1: Which emission sources have highest variability?

- **Answer:** **NonRenewable_Energy_Consumption_kWh** has the highest variability among the sources.
- **Justification:** Based on the "**Rolling Variance of Carbon Emission & Energy Sources**" plot, the green line representing non-renewable energy shows extreme fluctuations and reaches sharp variance peaks exceeding 1.5×10^9 , which is significantly higher than renewable energy.

Q2: Are emissions normally distributed?

- **Answer:** **No, the emissions are not normally distributed.** They follow a **Right-Skewed (Positively Skewed)** distribution.
- **Justification:** According to the "**Distribution of Carbon_Emission_tCO2e**" histogram, the data is heavily concentrated at the lower values (between 20 and 40) and stretches out with a long right tail towards higher emission values exceeding 140.

Q3: Which variables are strongly correlated?

- **Answer:** The strongest positive correlations are between:
 1. **Carbon_Emission_tCO2e** and **NonRenewable_Energy_Consumption_kWh**.
 2. **Total_Energy_Consumption_kWh** and **NonRenewable_Energy_Consumption_kWh**.
- **Justification:** The "**Correlation Heatmap**" indicates a very strong positive correlation coefficient of **0.87** between Carbon Emissions and Non-Renewable Energy. This linear relationship is also clearly verified by the upward-sloping pattern in the corresponding **Scatter Plot**.

Q4: Is there a trend in emissions?

- **Answer:** **Yes, there is a slight, gradual upward trend** over the long term.
- **Justification:** In the "**Time Series Decomposition - Trend**" plot, the trend component shows a continuous and steady increase from the beginning of 2024 (starting below 66.2) until the end of the year (reaching above 66.8). This gradual increase is also supported by the red **Moving Average** line.

Q5: Do seasonal patterns exist?

- **Answer: Yes, distinct and repetitive seasonal patterns exist** within the data.
- **Justification:** The "**Seasonality**" component from the time series decomposition displays a highly regular, cyclical wave pattern that repeats identically across different months. This constant, rhythmic fluctuation confirms the presence of clear seasonal behavior.